FACULTY OF ENGINEERING OF THE UNIVERSITY OF PORTO

Systems Engineering

Team Meeting Report

SWAID — Plate Resonance (PR)

Meeting Date:	March 4, 2026
Location:	FEUP I-108
Time:	16:10 — 17:45
Proposed by:	Professor Sérgio Reis Cunha

1 Attendance & Logistics

Team Member	Present	Role / Focus
Miguel Amorim	[P]	Team Leader
João Oliveira	[P]	Technical Leader (HW)
Artur Kovbasa	[P]	Technical Leader (SW)
Afonso Cruz	[P]	Member
Alexandre Gorra	[P]	Member
Beatriz Martins	[F]	Member
Hugo Gomes	[P]	Member
João Pinto	[P]	Member
Lucas Aparicio	[P]	Member

2 Meeting Objectives & Discussion Notes

2.1 Chladni Symbol Alphabet Methodology

Discussion & Notes:

- Establish an efficient methodology to define the ”alphabet” of symbols.
- Technical feasibility: Assess the cost-benefit of a custom simulator for resonant modes.
- Control variables: Material and geometry of the Chladni plate; quantity and positioning of transducers.

Conclusions:

- Validate the current simulator’s accuracy against known physical benchmarks.
- Automate simulations across various configurations to define the symbol alphabet.
- Experiment with diverse mathematical functions, plate geometries, and transducer counts.
- Coordinate with Team 12 to analyze candidate materials.
- Incorporate particle visualization (not just wave patterns) to better identify distinct symbols.
- Deepen theoretical understanding of the underlying physics and the simulator’s internal logic.
- Prioritize testing geometries that exhibit high modal density to ensure a rich variety of patterns.

2.2 Illumination Effects

Discussion & Notes:

- Brainstorm lighting ideas with cross-functional teams to enhance visual feedback.

Conclusions:

- **Negative Space Projection:** Projecting LEDs from beneath the plate. *Challenge:* Requires a transparent plate (e.g., acrylic); potential "dead zones" around transducers. Overhead laser projection was deemed too complex and costly.
- **Digital/AR Effects:** Visualizing the symbol digitally on a monitor, synchronized with the physical generation.
- **Ambient Lighting:** Side-mounted LED matrices acting as audio visualizers (e.g., volume bars).
- **Infinity Mirror Plate:** Using a polished metallic plate as a mirror combined with a one-way mirror top and peripheral LEDs to create a 3D "infinite" resonance effect.
- **Fluorescence:** Utilizing UV/Blacklight LEDs with fine fluorescent powders to make the Chladni patterns glow. PWM control can modulate brightness based on audio intensity.

2.3 FPGA Platform Selection & Software Environment

Discussion & Notes:

- Evaluate specific FPGA recommendations for signal processing workloads.
- Investigate open-source toolchains (e.g., Yosys) vs. proprietary suites.
- Define communication requirements and I/O channel estimates.

Conclusions:

- Using a dedicated sound card might simplify initial development compared to an FPGA.
- Perform a comparative analysis of sound card vs. FPGA solutions (latency, scalability, complexity).

2.4 Weekly Deliverable: Project Requirements

Discussion & Notes:

- Finalizing the "Project Requirements Document" for upcoming submission.
- Coordinate shared documentation sections with other teams (Global Vision).

Conclusions:

- Synchronize common topics with Teams 12 and 14 to ensure project cohesion.
- Assign a core group (4 members) to the document; others will focus on technical R&D.
- Clearly define the development boundaries and interfaces between the three sub-projects.

2.5 Team Workflow & Management Tools

Discussion & Notes:

- Evaluating OpenProject vs. GitHub for task management.
- Reminder regarding bi-weekly peer evaluations.

Conclusions:

- OpenProject was deemed unnecessarily complex for this specific scope.
- Establish a unified GitHub Organization for all three team repositories.
- Utilize GitHub Projects for task tracking and Kanban boards.
- Integrate GitHub with Discord for automated notifications (PRs, Commits, Deadlines).

3 Task Assignment

*Categories: Management (**G**), Deliverables (**E**), Development (**D**).*

Task Description	Cat.	Assigned To	Status	Deadline
Team Manual (Previous Task)	E	Entire Team	**Closed**	27/02
Project Requirements Document	E	Miguel, Lucas, Afonso, João Oliveira	Open	06/03
Complete Peer Evaluation Forms	G	Everyone	Open	07/03
Setup GitHub SWAID Organization (All teams)	G	Artur	Open	08/03
Configure Discord Server Integrations & Community (All teams)	G	Hugo	Open	08/03
Update Team Manual (New Tools/Org)	G	Miguel	Open	08/03
Hardware Platform Selection (FPGA vs. Sound Card)	D	Miguel, Lucas, Afonso, João O.	Open	11/03
Refine Chladni Simulator	D	Artur, Alexandre, Hugo	Open	11/03
Research Interactive Illumination Effects for Music Visualization & Hardware Specs	D	João Pinto, Beatriz	Open	11/03
Collaborate on Embedded Platform Selection	E	Entire Team / HI Team	Open	11/03
Setup GitHub Project Development Plan	G	Miguel, Artur, João O.	Open	13/03

4 Next Meeting

- **Proposed Date:** March 11, 2026
- **Proposed Location:** FEUP I-108

4.1 Next week's main objectives:

- Present a set of candidates configurations for the Chladni plate (number of transducers and their location)
- Solid light effects proposals to go along with the Chladni symbols
- Select a hardware platform (Sound Card or FPGA)
- Have a well though development plan
- Collaborative toolchain completly configured and in use